

John Doe

Instrumentation & Control Engineer

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PROFESSIONAL SUMMARY

Instrumentation & Control Engineer with 8+ years of experience designing, configuring, and commissioning control systems for energy infrastructure. Proven expertise in DCS, PLC, HMI, and SCADA systems across complex industrial facilities. Strong background in instrumentation detailed design including schematics, I/O drawings, wiring diagrams, and control panel layouts. Experienced in leading engineering deliverables from conceptual design through commissioning, mentoring junior engineers, and ensuring compliance with industry codes and standards.

CORE SKILLS

- Process instrumentation design and specification
- Instrument index and specification sheets (ISA data sheets)
- Hook-up drawings and installation details
- PLC programming (Allen-Bradley, Siemens S7)
- SIS design per IEC 61511 / ISA 84
- Instrument calibration and loop checking
- Smart transmitter configuration (Emerson, Yokogawa, Endress+Hauser)
- Project planning and execution
- FAT / SAT coordination
- Technical documentation and as-built records
- AutoCAD / MicroStation
- SAP PM module
- Functional safety lifecycle management
- SIL assessment and classification
- Gas facility instrumentation and control design
- Telemetry and remote SCADA systems
- MS-Project for project planning
- P&ID development and interpretation
- Loop diagrams and wiring design
- DCS configuration (Honeywell Experion, Emerson DeltaV)
- SCADA systems and HMI design
- SIL verification and validation
- HART / Foundation Fieldbus / Profibus protocols
- Calibration management (Beamex CMX, Fluke tools)
- Vendor technical evaluation and procurement
- Commissioning and start-up support
- SmartPlant Instrumentation (SPI / INtools)
- MATLAB / Simulink
- Microsoft Office Suite
- HAZOP and LOPA participation
- Compliance: IEC 61511, IEC 61508, ISA 84, API 554, NEC/NFPA
- Control and system functional narratives
- CSA codes and Canadian standards compliance

PROFESSIONAL EXPERIENCE

Senior Instrumentation Engineer

Jan 2021 - Present

Petronova Engineering Houston, TX

- Lead instrumentation engineering for greenfield and brownfield projects valued at \$50M+
- Develop instrument specifications, data sheets, and hook-up drawings for pressure, temperature, level, and flow instruments
- Design and review P&IDs, loop diagrams, and cable schedules using SmartPlant Instrumentation
- Configure and commission DCS (Honeywell Experion) and SIS (Triconex) for refinery units
- Conduct SIL verification studies and prepare safety requirement specifications (SRS)

- Coordinate with multidisciplinary teams (process, electrical, piping) during FEED and detailed design
- Mentor junior engineers and review deliverables for quality assurance

Instrumentation Engineer

Mar 2017 - Dec 2020

Gulf Coast Process Solutions Pasadena, TX

- Executed instrumentation detailed design for petrochemical and gas processing facilities
- Prepared instrument index, I/O lists, junction box layouts, and cable routing drawings
- Performed loop checking, calibration, and pre-commissioning of 2,000+ instrument loops
- Configured HART smart transmitters and control valves with digital positioners
- Supported DCS (Emerson DeltaV) point-to-point checkout and logic validation
- Participated in HAZOP reviews and implemented LOPA recommendations

Junior Instrumentation Engineer

Jun 2014 - Feb 2017

Meridian Industrial Services Baton Rouge, LA

- Assisted in design and installation of instrumentation for power plant upgrades
- Created instrument hook-up drawings and typical installation details in AutoCAD
- Performed field calibration of pressure, temperature, level, and flow transmitters
- Maintained calibration records and updated instrument database in SPI/INtools
- Supported plant turnaround activities including instrument inspection and replacement

EDUCATION

B.Sc. in Electrical Engineering (Instrumentation & Control)

2014

Louisiana State University Baton Rouge, LA

CERTIFICATIONS

- Certified Functional Safety Engineer (CFSE) - TUV Rheinland
- ISA Certified Control Systems Technician (CCST Level III)
- OSHA 30-Hour General Industry Safety

PROFESSIONAL MEMBERSHIPS

- International Society of Automation (ISA) - Senior Member
- Society of Petroleum Engineers (SPE)

LANGUAGES

- English - Native
- Spanish - Professional working proficiency